

Tanmay Kumar Sahu

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EDUCATION

New York University

M.S. in Computer Science, GPA: 3.8

Sep 2025 - May 2027

New York, NY

Coursework: Design and Analysis of Algorithms, Database Management Systems, Big Data, Machine Learning, Programming for Data Science, Information Visualization

Honors & Awards: Adobe Campus Ambassador; NVIDIA Spark New York Most Secure Project (MetaNemo, with Pensar); Baseten Healthcare AI Hackathon Winner; Intuit TechWeek NYC 2026 3rd Place; Dell x NVIDIA Hackathon (RiskTwin)

Training: CodePath Applications of AI Engineering (AI201, present); CodePath AI110 / Technical Interview Prep 101; NYU LeetCode Bootcamp

Amity University

B.Tech. in Computer Science and Engineering

Sep 2020 - Nov 2024

Uttar Pradesh, India

Honors & Awards: Team Lead, Amity AI Club; Tutor, Stanford Code in Place; IBM watsonX Hackathon Winner

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML/CSS, C, C++

Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Django REST Framework, Spring Boot, REST/GraphQL APIs, WebSockets, JWT, Zod

Infra / Data: Docker, Git/GitHub, Linux, PostgreSQL, Redis, MongoDB, MySQL, PostGIS, Azure workflows, Vercel, Supabase

Practices: API design, schema design, testing, debugging, Docker Compose, CI-friendly deploys, technical documentation

EXPERIENCE

NYU Tandon School of Engineering

VIP Research Team Member, Defining Accessible XR

Sep 2026 - Present

Brooklyn, NY

- Built sensor-driven guidance logic in a VR navigation system that leads a blind and deaf user home, detecting wrong-way movement and nearby obstacles and triggering corrective feedback in real time.
- Ran 20+ testing rounds with 5 test users so the team could iterate on guidance and obstacle alerts, then turned observations into fixes the team of nearly 20 members shipped into the prototype.
- Documented test protocols and findings for the Helen Keller Services collaboration, giving the research team repeatable evaluations and evidence-based design recommendations (ongoing).

GBCS Group - SkyIT

Backend Developer Intern

Jun 2026 - Aug 2026

Remote

- Built and hardened about 10 backend API endpoints for the company's fleet-management product using Django REST Framework, Express.js, REST, and GraphQL, so ops and frontend teams could pull vehicle and job data; 50+ people used the work.
- Fixed about 30 slow queries and bugs across PostgreSQL, MySQL, MongoDB, and Microsoft Azure workflows, making search operations and calculations noticeably faster and unblocking customer-facing endpoints.
- Cut integration friction by aligning API contracts, Git workflows, and Docker-based deploys with how product, frontend, and DevOps shipped features, and debugged failed deploys along the way.

UniConverge Technologies

Data Science and Machine Learning Intern

May 2023 - Jul 2023

Noida, India

- Delivered Decision Tree and Extra Trees models the client used for smart-city resource planning, improving planning accuracy by about 15-20% versus their prior baseline.
- Turned traffic and weather feeds into repeatable SQL/Pandas pipelines the team could retrain from, with MAE checks and feature work on date_time, vehicles, and junctions.

SELECTED PROJECTS

MetaNemo - Edge AI Wearable Platform

Node.js, FastAPI, Docker, PostgreSQL/PostGIS, WebSockets

- Owned the service path from Meta Ray-Ban clients to local NVIDIA DGX Spark inference: Node.js WebSocket middleware, FastAPI agent services, Docker Compose, and Tailscale/WireGuard tunnels.
- Added PostGIS/H3 storage and a nightly NYC Open Data ingest job so location features stayed fresh without paid cloud model APIs; Most Secure Project at NVIDIA Spark New York (with Pensar).

DataNuts - Full-Stack Learning Platform

TypeScript, Next.js, Node.js, PostgreSQL, JWT

- Shipped a practice product where students authenticate, solve topic/difficulty-tagged problems in an in-browser editor, and track progress with analytics and achievement profiles.
- Implemented JWT + bcrypt sessions, problem-repository APIs, and progress models so content could grow without rewriting core auth and submission flows.

Lazarus Mesh - Verified Recovery Marketplace

TypeScript, Node.js, PostgreSQL, Vercel

- Designed a local-first marketplace MVP with authenticated merchant flows, Ed25519-signed quotes, pinned manifests, per-piece and reconstructed-content hash checks, policy caps, and audit events.

Snickr - Collaboration Backend

Java, Spring Boot, PostgreSQL, Docker Compose

- Built a Slack-like API for users, workspaces, channels, messages, invitations, and search on a normalized PostgreSQL schema with bcrypt sessions, rate limiting, GIN full-text indexes, and Dockerized local setup.

GITHUB CONTRIBUTIONS AND RESEARCH

- GitHub:** Active builder at github.com/7dracoder; merged open-source contributor to Supafone Labs (fixed local test collection, refreshed live-capability examples and docs, prevented test credential leakage, and validated Python, MCP, provider-contract, and TypeScript suites).
- Research:** Springer (2024), Real Time Video Dehazing and Object Detection using Deep Learning; IEEE (2023), Real Time Video Object Detection using Deep Learning. Current VIP research at NYU Tandon on accessible XR with Helen Keller Services.